Diffusion Models III

Lecture 9: Stochastic differential equations (SDE) and continuous time diffusion models

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Continuous-time diffusion

Goal: connect stochastic paths, probability densities, and sampling.

$$dX_t = \underbrace{f(X_t, t)\, dt}_{\text{systematic motion}} + \underbrace{g(t)\, dW_t}_{\text{random fluctuation}}, \qquad X_t \sim q_t.$$

1. **What does an SDE mean?**
   Understand drift, Brownian noise, and simulation of sample paths.
2. **How does the SDE determine the density q_t?**
   The Fokker–Planck equation describes how probability moves.
3. **Why can Langevin dynamics sample a target density?**
   Study how drift and noise produce stationarity and convergence.
4. **How can an ODE generate the same marginal densities?**
   Match the SDE's probability flux to derive the probability-flow ODE.

Brownian motion

Standard Brownian motion, also called a **Wiener process**, is the basic continuous-time model of accumulated random noise.

- $W_0 = 0$ and paths are continuous.
- For $h > 0$,

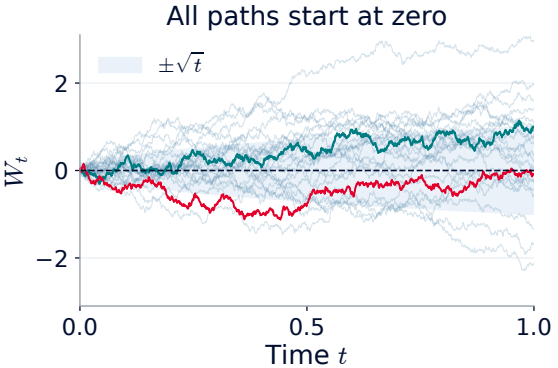
$$W_{t+h} - W_t \sim \mathcal{N}(0, h).$$

- Increments over disjoint intervals are independent.

Consequently,

$$\mathbb{E}[W_t] = 0, \qquad \mathrm{Var}(W_t) = t.$$

The typical spread grows as $\sqrt{t}$.



Brownian increments

Let W_t be a standard one-dimensional Brownian motion.

Finite increments. For an interval of length $h > 0$,

$$\Delta W_t := W_{t+h} - W_t \sim \mathcal{N}(0, h), \qquad \mathbb{E}[\Delta W_t] = 0, \qquad \mathrm{Var}(\Delta W_t) = h.$$

The increment is independent of the past. Its standard deviation is $\sqrt{h}$.

Differential notation. The symbol dW_t is understood through stochastic integration:

$$\boxed{\int_a^b dW_t = W_b - W_a.}$$

Why ordinary differentiation fails? The finite-difference slope satisfies

$$\frac{W_{t+h} - W_t}{h} \sim \mathcal{N}\left(0, \frac{1}{h}\right), \qquad \lim_{h \to 0} \mathrm{Var}\left(\frac{W_{t+h} - W_t}{h}\right) = \lim_{h \to 0} \frac{1}{h} = \infty.$$

These Gaussian slopes do not converge to a finite derivative. In fact, **Brownian paths are almost surely nowhere differentiable.**

Brownian quadratic variation

Fix $t > 0$ and partition $[0, t]$ into n equal intervals of length $h = t/n$, with endpoints $t_k = kh$ for $k = 0, \ldots, n$. Define the **quadratic covariation** and **quadratic variation**:

$$[U, V]_t := \lim_{n\to\infty} \sum_{k=0}^{n-1} \left(U_{t_{k+1}} - U_{t_k}\right)\left(V_{t_{k+1}} - V_{t_k}\right),$$

$$[U, U]_t := \lim_{n\to\infty} \sum_{k=0}^{n-1} \left(U_{t_{k+1}} - U_{t_k}\right)^2.$$

where the limit is taken **in probability**, when it exists.

Differential-product notation. We define

$$\boxed{dU_t\, dV_t := d[U, V]_t, \qquad (dU_t)^2 := d[U, U]_t.}$$

Its integrated meaning is

$$\int_0^t dU_s\, dV_s = [U, V]_t, \qquad \int_0^t (dU_s)^2 = [U, U]_t.$$

Brownian multiplication rules

Brownian quadratic variation. Independent $\mathcal{N}(0, h)$ increments give

$$Q_n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2, \qquad \mathbb{E}[Q_n] = t, \quad \mathrm{Var}(Q_n) = \frac{2t^2}{n}.$$

By Markov's inequality, $Q_n \xrightarrow{\mathbb{P}} t$, so $[W, W]_t = t$.

Products involving time.

$$\sum_{k=0}^{n-1} (t_{k+1} - t_k)^2 = t^2/n \to 0,$$

$$\sum_{k=0}^{n-1} (t_{k+1} - t_k)(W_{t_{k+1}} - W_{t_k}) = h(W_t - W_0) \xrightarrow{\mathbb{P}} 0.$$

By the definition of differential products,

$$\boxed{(dW_t)^2 = dt, \qquad dt\, dW_t = 0, \qquad (dt)^2 = 0.}$$

Itô's chain rule

For a twice continuously differentiable function ϕ, Itô's formula gives

$$d\phi(W_t) = \phi'(W_t)\, dW_t + \underbrace{\frac{1}{2}\phi''(W_t)\, dt}_{\text{Itô correction}}.$$

The correction arises from the **second-order Taylor term**. Writing $\Delta W_k = W_{t_{k+1}} - W_{t_k}$,

$$\phi(W_{t_{k+1}}) - \phi(W_{t_k}) = \phi'(W_{t_k})\Delta W_k + \frac{1}{2}\phi''(W_{t_k})(\Delta W_k)^2 + R_k.$$

After summing over the partition and taking the limit, the accumulated remainder vanishes. The quadratic term survives: the rule $(dW_t)^2 = dt$ gives its contribution $\frac{1}{2}\phi''(W_t)\, dt$.

The Itô integral

Brownian paths are not differentiable, so $\int H_t\, dW_t$ needs its own definition. Assume H_t is adapted and continuous in mean square on $[0, T]$, with $\mathbb{E}[H_t^2] < \infty$. For equally spaced times $t_k = kT/n$, use **left endpoints**:

$$I_T := \int_0^T H_t\, dW_t = \lim_{n\to\infty} \sum_{k=0}^{n-1} H_{t_k}\left(W_{t_{k+1}} - W_{t_k}\right).$$

- H_t may be **deterministic or random** (e.g., $H_t = e^{-t}$ or $H_t = W_t$.)
- H_t uses only information available up to time t, including the current value W_t.
- If $\mathbb{E}[\int_0^T H_t^2\, dt] < \infty$, then

$$\mathbb{E}[I_T] = 0, \qquad \mathbb{E}[I_T^2] = \mathbb{E}\left[\int_0^T H_t^2\, dt\right].$$

The second identity is **Itô's isometry**; it also gives $\mathrm{Var}(I_T)$.

The Itô SDE

An **Itô stochastic differential equation** describes a state $X_t \in \mathbb{R}^D$ driven by standard Brownian motion $W_t \in \mathbb{R}^m$:

$$\boxed{dX_t = \underbrace{f(X_t, t)\, dt}_{\text{systematic drift}} + \underbrace{G(X_t, t)\, dW_t}_{\text{Brownian noise}}.}$$

Here $f(x, t) \in \mathbb{R}^D$ specifies the drift, while $G(x, t) \in \mathbb{R}^{D \times m}$ scales and mixes the noise. The local noise covariance per unit time is $a(x, t) = G(x, t)G(x, t)^\top$.

Given an initial state X_0, the SDE is shorthand for the **integral equation**

$$X_t = X_0 + \underbrace{\int_0^t f(X_u, u)\, du}_{\text{ordinary time integral}} + \underbrace{\int_0^t G(X_u, u)\, dW_u}_{\text{Itô integral}}.$$

Itô convention: the stochastic integral uses left-endpoint coefficients, determined *before* the next Brownian increment.

Itô's formula: one dimension

Consider the scalar SDE $dX_t = f\,dt + G\,dW_t$, and write $a = G^2$. For this heuristic derivation, assume $\phi(x,t)$ is twice continuously differentiable jointly in (x,t). **Itô's formula** gives

$$\boxed{d\phi(X_t,t) = \left[\partial_t\phi + f\,\partial_x\phi + \frac{1}{2}a\,\partial_{xx}\phi\right]dt + G\,\partial_x\phi\,dW_t.}$$

where $\partial_t\phi = \frac{\partial\phi(x,t)}{\partial t}$, $\partial_x\phi = \frac{\partial\phi(x,t)}{\partial x}$, $\partial_{xx}\phi = \frac{\partial^2\phi(x,t)}{\partial x^2}$.

Derivation. By the second-order Taylor expansion,

$$d\phi(X_t,t) = \partial_t\phi\,dt + \partial_x\phi\,dX_t + \frac{1}{2}\partial_{xx}\phi\,(dX_t)^2 + \frac{1}{2}\partial_{tt}\phi\,\underbrace{(dt)^2}_{=0} + \partial_{xt}\phi\,\underbrace{dt\,dX_t}_{=0}.$$

The Itô multiplication rules give

$$(dX_t)^2 = f^2\,\underbrace{(dt)^2}_{=0} + 2fG\,\underbrace{dt\,dW_t}_{=0} + G^2\,\underbrace{(dW_t)^2}_{=dt} = a\,dt.$$

Substituting dX_t and $(dX_t)^2$ to the second-order Taylor expansion yields the result.

Itô's formula: general case

Let $X_t \in \mathbb{R}^D$ be driven by standard Brownian motion $W_t \in \mathbb{R}^m$:

$$dX_t = f\,dt + G\,dW_t, \qquad f \in \mathbb{R}^D, \quad G \in \mathbb{R}^{D\times m}, \quad a = GG^\top.$$

For a scalar function $\phi(x,t) \in C^{1,2}$, **Itô's formula** (also called **Itô's lemma**) is

$$\boxed{d\phi(X_t,t) = \left[\partial_t\phi + f^\top \nabla_x \phi + \frac{1}{2}\,\mathrm{tr}\left(a\nabla_x^2\phi\right)\right] dt + (\nabla_x\phi)^\top G\,dW_t.}$$

Here $\nabla_x\phi$ is the gradient and $\nabla_x^2\phi$ is the Hessian. The correction combines the local covariance with the curvature of ϕ:

$$dX_t\,dX_t^\top = a\,dt, \qquad \mathrm{tr}\left(a\nabla_x^2\phi\right) = \sum_{i,j=1}^{D} a_{ij}\,\partial_{x_i x_j}\phi.$$

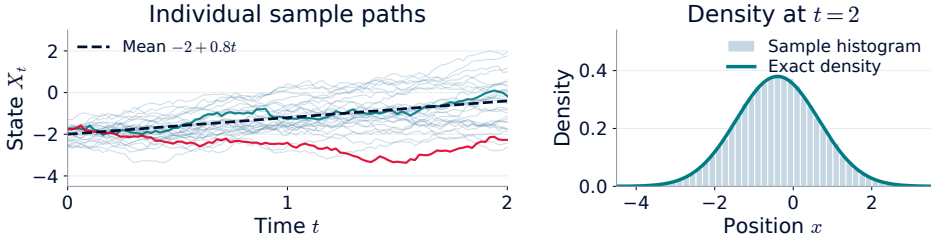
Simulating an Itô SDE

Choose a step size $h > 0$ and set $t_k = kh$. Let x_k approximate the state X_{t_k}. The **Euler–Maruyama update** is

$$\boxed{x_{k+1} = x_k + h\, f(x_k, t_k) + \sqrt{h}\, G(x_k, t_k) z_k} \qquad z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_m).$$

Use the current coefficients and a fresh Brownian increment $\sqrt{h}\, z_k$.

Example. $dX_t = 0.8\, dt + 0.7\, dW_t$, with $X_0 \sim \mathcal{N}(-2, 0.35^2)$.

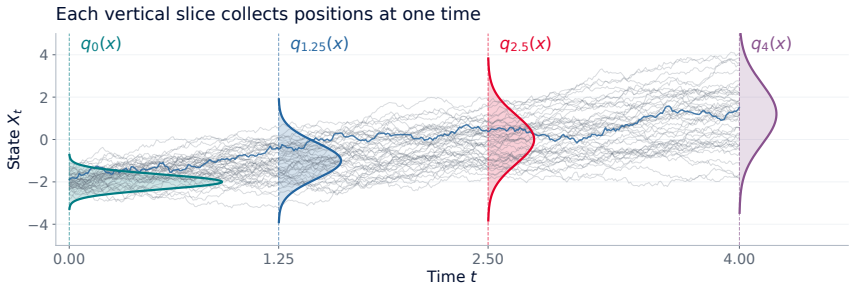


40,000 simulated paths (30 shown), $h = 0.02$. Histogram: endpoints at $t = 2$; curve: exact marginal density.

Paths and marginals

For $dX_t = 0.8\,dt + 0.7\,dW_t$, with $X_0 \sim \mathcal{N}(-2, 0.35^2)$,

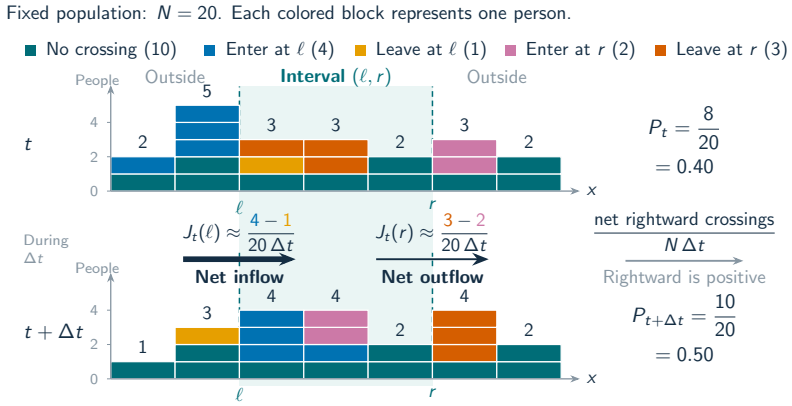
$$\boxed{X_t \sim \mathcal{N}\Big(-2 + 0.8t,\ 0.35^2 + 0.7^2 t\Big).}$$



A **path** follows one realization through time.
A **marginal density** $q_t(x)$ describes the distribution at a fixed time.

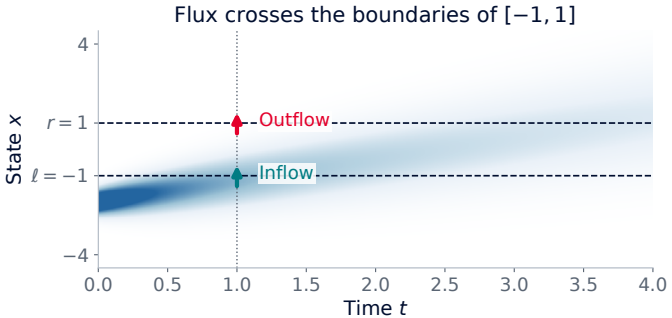
Probability flux

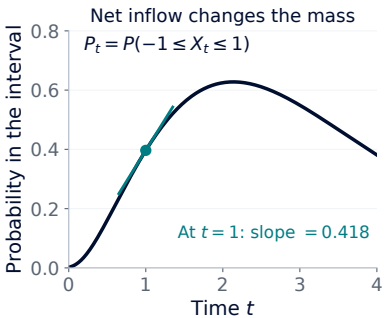
The **probability flux** $J_t(x)$ measures the **net rate of probability flow across** x. Rightward flow is positive; leftward flow is negative.



Flux across time

For a fixed interval, $P_t = P(\ell \le X_t \le r)$ changes at rate $\frac{d}{dt} P_t = J_t(\ell) - J_t(r)$.





Positive flux points **upward**, because the state axis is vertical.

Open animation in browser

Probability flux properties

By the definition of probability flux: For a fixed interval (ℓ, r),

$$\underbrace{\frac{d}{dt}\int_{\ell}^{r} q_t(x)\,dx}_{\Pr(\ell<X_t<r)} = \underbrace{J_t(\ell)}_{\text{signed inflow at left}} - \underbrace{J_t(r)}_{\text{signed outflow at right}}.$$

Let $F_t(x) = \Pr(X_t \le x)$. Taking the interval to be $(-\infty, x]$, with vanishing flux at $-\infty$, gives

$$\boxed{\partial_t F_t(x) = -J_t(x).}$$

Positive flux across x decreases the probability to its left.

Consider a small bin $(x, x+h)$ with $h \to 0$:

$$\underbrace{\frac{1}{h}\int_{x}^{x+h} \partial_t q_t(u)\,du}_{\longrightarrow\, \partial_t q_t(x)} = -\underbrace{\frac{J_t(x+h) - J_t(x)}{h}}_{\longrightarrow\, \partial_x J_t(x)}.$$

This gives the **continuity equation**

$$\boxed{\partial_t q_t(x) = -\partial_x J_t(x).}$$

Continuity equation in $\mathbb{R}^D$

For $x \in \mathbb{R}^D$, let $J_t(x) \in \mathbb{R}^D$ be the probability flux. For any fixed bounded region V with smooth boundary ∂V, let $n(x)$ be its **outward unit normal**.

Probability conservation and the **divergence theorem** give

$$\frac{d}{dt} \underbrace{\int_V q_t(x)\, dx}_{\Pr(X_t \in V)} = - \underbrace{\int_{\partial V} J_t(x) \cdot n(x)\, dS}_{\text{net outward probability flux}}$$

$$\overset{\text{divergence theorem}}{=} - \int_V \nabla \cdot J_t(x)\, dx.$$

Since this holds for every V, sufficient smoothness gives the **continuity equation**:

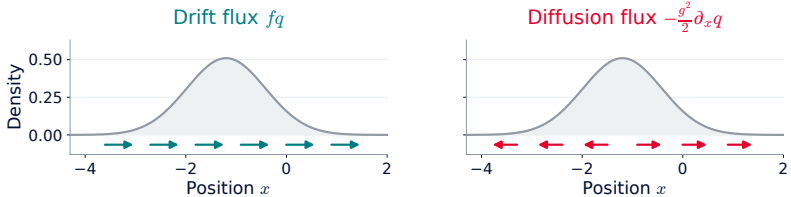
$$\partial_t q_t(x) = -\nabla \cdot J_t(x) = -\sum_{i=1}^{D} \frac{\partial J_{t,i}(x)}{\partial x_i}.$$

- Internal transfers cancel. Only flow across ∂V changes the probability inside V.
- $\nabla \cdot J_t(x) > 0$ means **local net outflow**, so the density $q_t(x)$ decreases.

The Fokker–Planck equation

For $dX_t = f(X_t, t)\,dt + g(t)\,dW_t$, the **Fokker–Planck equation** shows that the flux has two parts:

$$J_t(x) = f(x,t)q_t(x) - \frac{g(t)^2}{2}\partial_x q_t(x).$$



Substitute into $\partial_t q_t = -\partial_x J_t$:

$$\boxed{\partial_t q_t(x) = -\partial_x(f(x,t)q_t(x)) + \frac{g(t)^2}{2}\partial_x^2 q_t(x).}$$

Prove Fokker–Planck equation

Consider the one-dimensional Itô SDE

$$dX_t = f(X_t, t)\,dt + G(X_t, t)\,dW_t, \qquad a(x,t) = G(x,t)^2.$$

Let q_t be the density of X_t. Choose any smooth function $\phi(x)$ that is zero outside a bounded interval. **Itô's formula** gives

$$d\phi(X_t) = \left[f(X_t, t)\phi'(X_t) + \frac{1}{2} a(X_t, t)\phi''(X_t) \right] dt + G(X_t, t)\phi'(X_t)\,dW_t.$$

The Itô integral has mean zero. Taking expectations gives

$$d\mathbb{E}\phi(X_t) = \mathbb{E}\left[f(X_t, t)\phi'(X_t) + \frac{1}{2} a(X_t, t)\phi''(X_t) \right] dt.$$

Moving dt to the left gives

$$\boxed{\int_{\mathbb{R}} \phi\, \partial_t q_t\, dx = \int_{\mathbb{R}} f q_t\, \phi'\, dx + \frac{1}{2} \int_{\mathbb{R}} a q_t\, \phi''\, dx.}$$

Proof continued

Integrate by parts; the compact support of ϕ removes boundary terms:

$$\int_{\mathbb{R}} f q_t \, \phi' \, dx = -\int_{\mathbb{R}} \phi \, \partial_x (f q_t) \, dx,$$

$$\int_{\mathbb{R}} a q_t \, \phi'' \, dx = -\int_{\mathbb{R}} \phi' \, \partial_x (a q_t) \, dx$$

$$= \int_{\mathbb{R}} \phi \, \partial_x^2 (a q_t) \, dx.$$

Substituting gives

$$\int_{\mathbb{R}} \phi \left[\partial_t q_t + \partial_x (f q_t) - \frac{1}{2} \partial_x^2 (a q_t) \right] dx = 0.$$

Since ϕ is arbitrary, we obtain the **Fokker–Planck equation**:

$$\boxed{\partial_t q_t(x) = -\partial_x [f(x,t) q_t(x)] + \frac{1}{2} \partial_x^2 [a(x,t) q_t(x)] \, .}$$

Identifying the probability flux

Write the density equation in **conservation form**:

$$\partial_t q_t = -\partial_x \left[f q_t - \frac{1}{2} \partial_x (a q_t) \right].$$

Comparing with $\partial_t q_t = -\partial_x J_t$, the associated flux is

$$\boxed{J_t(x) = f(x,t) q_t(x) - \frac{1}{2} \partial_x [a(x,t) q_t(x)].}$$

If $G(x,t) = g(t)$ is independent of position, the flux simplifies to

$$J_t(x) = f(x,t) q_t(x) - \frac{1}{2} g(t)^2 \, \partial_x q_t(x).$$

- The **drift flux** $f q_t$ has direction determined by f.
- The **diffusion flux** $-\frac{1}{2} g^2 \partial_x (q_t)$ points from higher toward lower density.

Together, the two contributions determine the net inflow into each interval.

Fokker–Planck in $\mathbb{R}^D$

For the general Itô SDE,

$$dX_t = f(X_t, t)\,dt + G(X_t, t)\,dW_t, \qquad a(x,t) = G(x,t)G(x,t)^\top,$$

where $X_t, f \in \mathbb{R}^D$, $W_t \in \mathbb{R}^m$, and $G \in \mathbb{R}^{D\times m}$, the same argument gives

$$\boxed{\partial_t q_t(x) = -\sum_{i=1}^{D} \partial_{x_i}[f_i(x,t)q_t(x)] + \frac{1}{2}\sum_{i,j=1}^{D} \partial_{x_i}\partial_{x_j}[a_{ij}(x,t)q_t(x)]\,.}$$

By the continuity equation in $\mathbb{R}^D$:

$$\partial_t q_t = -\nabla_x \cdot J_t,$$

we have

$$\boxed{J_{t,i}(x) = f_i(x,t)q_t(x) - \frac{1}{2}\sum_{j=1}^{D} \partial_{x_j}[a_{ij}(x,t)q_t(x)]\,, \quad i = 1,\dots,D.}$$

Together with the initial density q_0, this PDE describes how the marginal density evolves.

Langevin equilibrium

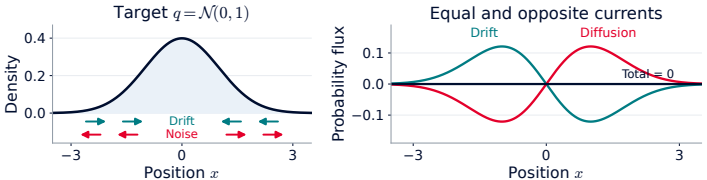
Consider the **Langevin SDE** $dX_t = \frac{1}{2}\nabla_x \log q(X_t)\, dt + dW_t$, where q is a smooth, positive target density.

Let q_t denote the current density of X_t. The Fokker–Planck equation gives the probability flux

$$J_t(x) = \frac{1}{2} q_t(x) \nabla_x \log q(x) - \frac{1}{2} \nabla_x q_t(x).$$

If $q_t = q$, then $q\nabla_x \log q = \nabla_x q$, so the **drift and diffusion fluxes cancel**:

$$\boxed{J_t(x) \equiv 0 \quad \Longrightarrow \quad \partial_t q_t(x) \equiv 0.}$$

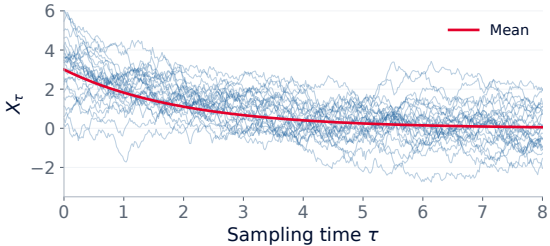


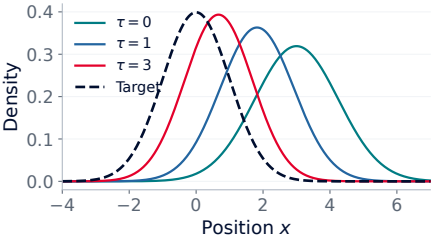
The density stays fixed while individual paths keep moving.

Langevin convergence

- **Stationarity:** If $X_0 \sim q$, then $X_\tau \sim q$ for every $\tau \geq 0$.
- **Convergence:** Under suitable ergodicity conditions, the distribution of X_τ converges to q as $\tau \to \infty$, even when X_0 has a different distribution.

Example: Consider $q = \mathcal{N}(0, 1)$, starting point $X_0 \sim \mathcal{N}(3, 1.25^2)$. The mean tends to 0 and variance to 1.





Langevin dynamics

Let $s(x) = \nabla_x \log q(x)$ be the score of the target density q. The **Langevin SDE** satisfies

$$dX_t = \frac{1}{2} s(X_t)\, dt + dW_t,$$

where W_t is D-dimensional Brownian motion and t is sampling time.

Integrating over $[t, t + \Delta t]$ gives

$$X_{t+\Delta t} = X_t + \frac{1}{2} \int_t^{t+\Delta t} s(X_u)\, du + \underbrace{W_{t+\Delta t} - W_t}_{\sim \mathcal{N}(0, \Delta t\, I_D)}$$

$$\approx X_t + \frac{\Delta t}{2} s(X_t) + \sqrt{\Delta t}\, Z,$$

where $Z \sim \mathcal{N}(0, I_D)$ is independent of X_t.

The approximation is an **Euler–Maruyama step**: hold the drift fixed at $s(X_t)/2$ throughout the interval. Repeating this update with fresh independent Gaussian noise gives the **discrete Langevin sampler**.

DDPM in continuous time

Rescale the DDPM step. On the grid $t_k = kh \in [0, 1]$, use step variance $\beta(t_k)h$:

$$x_{k+1} = \sqrt{1 - \beta(t_k)h}\, x_k + \sqrt{\beta(t_k)h}\, z_k, \qquad z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_D).$$

Here $x_k \approx X_{t_k}$, and $\beta(t) > 0$ is the noise rate.

Using $\sqrt{1-u} = 1 - \frac{1}{2}u + O(u^2)$ gives

$$x_{k+1} - x_k \approx -\frac{1}{2}\beta(t_k)x_k\, h + \sqrt{\beta(t_k)}\, \underbrace{\sqrt{h}\, z_k}_{\Delta W_k}.$$

Identify $\Delta W_k = W_{t_{k+1}} - W_{t_k}$ as a Brownian increment, with $\Delta W_k \sim \mathcal{N}(0, hI_D)$.

This first-order update is **Euler–Maruyama** for the **variance-preserving (VP) SDE**:

$$\boxed{dX_t = -\frac{1}{2}\beta(t)X_t\, dt + \sqrt{\beta(t)}\, dW_t.}$$

Solving the VP SDE

Consider the DDPM SDE:

$$dX_t = -\frac{1}{2}\beta(t)X_t\,dt + \sqrt{\beta(t)}\,dW_t, \qquad X_0 = x_0.$$

For a deterministic differentiable $c(t)$, **Itô's formula** gives

$$\begin{aligned} d(c(t)X_t) &= c(t)\,dX_t + c'(t)X_t\,dt \\ &= \left[c'(t) - \frac{1}{2}\beta(t)c(t)\right] X_t\,dt + c(t)\sqrt{\beta(t)}\,dW_t. \end{aligned}$$

Cancel the drift: choose $c'(t) = \frac{1}{2}\beta(t)c(t)$ and $c(0) = 1$. Then

$$c(t) = \exp\left[\frac{1}{2}\int_0^t \beta(u)\,du\right] = \bar{\alpha}(t)^{-1/2}, \qquad \bar{\alpha}(t) := \exp\left[-\int_0^t \beta(u)\,du\right].$$

Now $d(c(t)X_t) = c(t)\sqrt{\beta(t)}\,dW_t$. Integrating from 0 to t gives

$$\boxed{c(t)X_t = x_0 + \int_0^t c(u)\sqrt{\beta(u)}\,dW_u.}$$

The VP forward marginal

Dividing by $c(t) = \bar{\alpha}(t)^{-1/2}$ gives the solution

$$X_t = \sqrt{\bar{\alpha}(t)}\, x_0 + \sqrt{\bar{\alpha}(t)} \int_0^t \sqrt{\frac{\beta(u)}{\bar{\alpha}(u)}}\, dW_u.$$

This **Itô integral** is **Gaussian with mean zero** because its integrand is deterministic. Thus $\mathbb{E}[X_t \mid X_0 = x_0] = \sqrt{\bar{\alpha}(t)}\, x_0$.
Since $\frac{d}{du}\bar{\alpha}(u)^{-1} = \beta(u)/\bar{\alpha}(u)$, **Itô's isometry** gives

$$\begin{aligned} \mathrm{Cov}(X_t \mid X_0 = x_0) &= \bar{\alpha}(t) \int_0^t \frac{\beta(u)}{\bar{\alpha}(u)}\, du\, I_D \\ &= \bar{\alpha}(t) \left[\frac{1}{\bar{\alpha}(t)} - 1 \right] I_D = [1 - \bar{\alpha}(t)]\, I_D. \end{aligned}$$

$$\boxed{q(x_t \mid x_0) = \mathcal{N}\Big(x_t;\, \sqrt{\bar{\alpha}(t)}\, x_0,\, [1 - \bar{\alpha}(t)]\, I_D \Big).}$$

SMLD: the forward noising chain

SMLD (score matching with Langevin dynamics) uses noise levels $0 = \sigma_0 < \sigma_1 < \cdots < \sigma_L$, with perturbation marginals

$$q(x_k \mid x_0) = \mathcal{N}\Big(x_k;\, x_0,\, \sigma_k^2 I_D\Big)\,.$$

Connect adjacent levels by adding **independent Gaussian noise** with covariance $v_k I_D$. Matching the accumulated variance requires

$$\sigma_k^2 + v_k = \sigma_{k+1}^2 \quad \Longrightarrow \quad v_k = \sigma_{k+1}^2 - \sigma_k^2.$$

Thus a compatible **forward transition** is

$$q(x_{k+1} \mid x_k) = \mathcal{N}\Big(x_{k+1};\, x_k,\, \Big[\sigma_{k+1}^2 - \sigma_k^2\Big]\, I_D\Big)\,,$$

or equivalently,

$$\boxed{x_{k+1} = x_k + \sqrt{\sigma_{k+1}^2 - \sigma_k^2}\, z_k, \qquad z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_D).}$$

From SMLD to the VE SDE

Let $t_k = kh$, $\sigma_k = \sigma(t_k)$, and $\sigma^2(0) = 0$. A first-order expansion gives

$$x_{k+1} - x_k \approx \sqrt{(\sigma^2)'(t_k)}\,\Delta W_k, \qquad \Delta W_k = \sqrt{h}\,z_k.$$

The continuous-time limit is the **variance-exploding (VE) SDE**:

$$\boxed{dX_t = \sqrt{(\sigma^2)'(t)}\,dW_t.}$$

Given $X_0 = x_0$, integration gives

$$X_t = x_0 + \int_0^t \sqrt{(\sigma^2)'(u)}\,dW_u.$$

The integral is centered Gaussian, with covariance $\left[\int_0^t (\sigma^2)'(u)\,du\right] I_D = \sigma^2(t) I_D$. Hence

$$\boxed{q(x_t \mid x_0) = \mathcal{N}\Big(x_t;\, x_0,\, \sigma^2(t) I_D\Big)\,.}$$

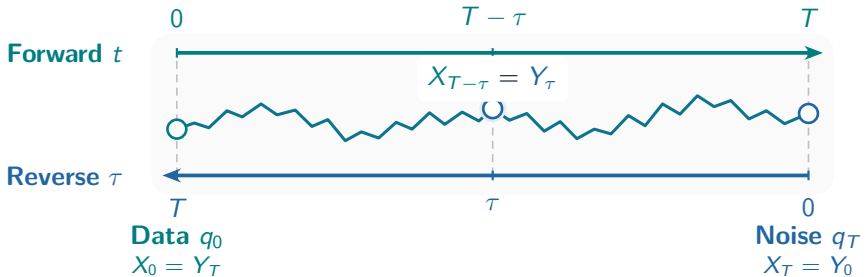
Reversing the clock

The forward SDE runs from data to noise:

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t, \qquad 0 \le t \le T.$$

Use an **increasing reverse clock** $\tau = T - t$:

$$\boxed{Y_\tau = X_{T-\tau}}, \qquad r_\tau(x) = q_{T-\tau}(x).$$



Reversing the drift alone is insufficient: the reverse drift also depends on the current density.

A constant-coefficient example

Goal: Show the **reverse process** is also **an Itô SDE**.

Start with a one-dimensional SDE with constant drift and noise scale:

$$\boxed{dX_t = \mu\, dt + \sigma\, dW_t}, \qquad X_t \sim q_t, \quad \sigma > 0.$$

For $h > 0$, the **forward transition is exact**:

$$X_t \mid X_{t-h} = y \sim N\Big(y + \mu h,\, \sigma^2 h\Big).$$

Define the **reverse increment**

$$\delta = X_{t-h} - X_t.$$

Given the current state $X_t = x$, by Bayes' rule, we have

$$p(\delta \mid X_t = x) \propto \underbrace{\exp\left[-\frac{(\delta + \mu h)^2}{2\sigma^2 h}\right]}_{\text{forward transition}} \underbrace{q_{t-h}(x + \delta)}_{\text{earlier density}}.$$

The score shifts the reverse mean

Let $s_t(x) = \partial_x \log q_t(x)$. Brownian increments scale as $\sqrt{h}$, so

$$\delta = O_p(\sqrt{h}), \qquad \delta^2 = O_p(h).$$

Expanding in x, then using time smoothness $s_{t-h}(x) = s_t(x) + O(h)$, gives

$$\begin{aligned}\log q_{t-h}(x+\delta) &= \underbrace{\log q_{t-h}(x)}_{\text{constant in } \delta} + s_{t-h}(x)\delta + O_p(h) \\ &= \log q_{t-h}(x) + s_t(x)\delta + O_p(h).\end{aligned}$$

Substitute into **Bayes' formula** and complete the square:

$$\begin{aligned}\log p(\delta \mid X_t = x) &\approx C - \frac{(\delta + \mu h)^2}{2\sigma^2 h} + s_t(x)\delta \\ &= C' - \frac{\{\delta - [-\mu + \sigma^2 s_t(x)]h\}^2}{2\sigma^2 h}.\end{aligned}$$

Thus:

$$\boxed{\delta \mid X_t = x \;\approx\; N\Big([-\mu + \sigma^2 s_t(x)]h,\; \sigma^2 h\Big).}$$

From the reverse step to an SDE

The local Gaussian approximation gives, for $h > 0$,

$$X_{t-h} \approx X_t + \left[-\mu + \sigma^2 s_t(X_t)\right] h + \sigma\sqrt{h}\, Z, \qquad Z \sim N(0,1).$$

Introduce an **increasing reverse clock**:

$$\tau = T - t, \qquad Y_\tau = X_{T-\tau}, \qquad Y_0 \sim q_T.$$

The backward step $t \to t - h$ becomes the forward step $\tau \to \tau + h$:

$$\boxed{Y_{\tau+h} \approx Y_\tau + \left[-\mu + \sigma^2 s_{T-\tau}(Y_\tau)\right] h + \sigma\sqrt{h}\, Z.}$$

This has the **Euler–Maruyama form** of

$$\boxed{dY_\tau = \left[-\mu + \sigma^2 \partial_x \log q_{T-\tau}(Y_\tau)\right] d\tau + \sigma\, dB_\tau,}$$

where B_τ is Brownian motion with respect to the reversed filtration.

- Under suitable regularity conditions, the **time-reversal theorem** of Anderson (1982) rigorously establishes this reverse Itô representation.

General reverse SDE

Consider the general forward SDE and its time reversal:

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t, \qquad Y_\tau = X_{T-\tau}.$$

Under suitable regularity conditions, the time-reversal theorem gives an Itô representation

$$dY_\tau = b(Y_\tau, \tau)\,d\tau + \gamma(\tau)\,dB_\tau.$$

Reversing a path preserves its squared increments:

$$(Y_{\tau_i} - Y_{\tau_{i-1}})^2 = (X_{T-\tau_{i-1}} - X_{T-\tau_i})^2.$$

Summing and taking the limit gives the **quadratic variation**

$$\boxed{[Y]_\tau = [X]_T - [X]_{T-\tau} = \int_{T-\tau}^{T} g^2(u)\,du = \int_0^\tau g^2(T-v)\,dv.}$$

The reverse SDE also gives

$$[Y]_\tau = \int_0^\tau \gamma(v)^2\,dv.$$

Comparing the two expressions gives: $\boxed{\gamma(\tau) = g(T-\tau)}$.

The drift of the reverse SDE

At matched times $t = T - \tau$, write the **marginal density of the reversed process** $r_\tau(x) = q_t(x)$. The forward **Fokker–Planck equation (FPE)** and flux are

$$\partial_t q_t = -\partial_x J_t, \qquad J_t = f(x,t)q_t - \frac{1}{2}g^2(t)\partial_x q_t.$$

Denote the **reverse flux** as $\widetilde{J}_\tau$. Reversing the clock gives

$$\partial_\tau r_\tau = -\partial_t q_t = \partial_x J_t = -\partial_x \widetilde{J}_\tau.$$

With vanishing flux at infinity, $\boxed{\widetilde{J}_\tau = -J_t}$. According to the **reverse FPE**:

$$\underbrace{b(x,\tau)r_\tau - \frac{1}{2}\gamma^2(\tau)\partial_x r_\tau}_{\text{reverse flux}} = \underbrace{b(x,\tau)q_t - \frac{1}{2}g^2(t)\partial_x q_t}_{\text{reverse flux equivalent form}} = \underbrace{-f(x,t)q_t + \frac{1}{2}g^2(t)\partial_x q_t}_{\text{negative forward flux}}.$$

Solving for b and dividing by $q_t > 0$ gives

$$\boxed{b(x,\tau) = -f(x,t) + g^2(t)\frac{\partial_x q_t(x)}{q_t(x)} = -f(x,t) + g^2(t)\partial_x \log q_t(x).}$$

The form of reverse SDE

With the **increasing reverse clock** τ, we obtained

$$dY_\tau = \Big[-f(Y_\tau, T-\tau) + g^2(T-\tau)\partial_x \log q_{T-\tau}(Y_\tau)\Big]\, d\tau + g(T-\tau)\, dB_\tau.$$

Return to the original time coordinate:

$$t = T - \tau, \qquad d\tau = -dt, \qquad Y_\tau = X_t.$$

Now t runs **backward from** T **to** 0, and the drift term becomes

$$\Big[-f + g^2 \partial_x \log q_t\Big]\, d\tau = \Big[f - g^2 \partial_x \log q_t\Big]\, dt.$$

Denoting the **reverse-time Brownian motion** by $\overline{W}_t$, we recover the form used in Song et al. (2021):

$$\boxed{dX_t = \Big[f(X_t, t) - g^2(t)\partial_x \log q_t(X_t)\Big]\, dt + g(t)\, d\overline{W}_t, \qquad dt < 0.}$$

In $\mathbb{R}^D$: replace ∂_x by ∇_x, with vector-valued states and drift, and D-dimensional Brownian motion.

The velocity field of an ODE

An **ordinary differential equation (ODE)** specifies a particle's rate of change:

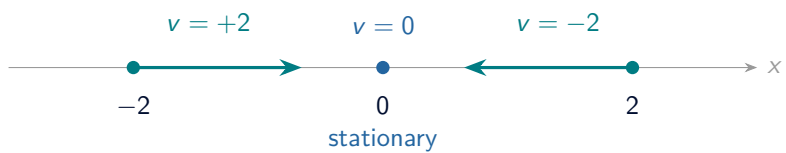
$$\boxed{\frac{dZ_t}{dt} = v(Z_t, t).}$$

The **velocity field** $v(x, t)$ gives the velocity at position x and time t. A particle evaluates this field at its current position, $x = Z_t$.

$$Z_{t+h} = Z_t + v(Z_t, t)h + o(h).$$

The sign of v gives direction; $|v|$ gives speed.

Example: $v(x, t) = -x$ moves particles toward zero.



The evolving density of ODE

Start many particles from $Z_0 \sim \rho_0$ and move each with the same velocity field. The density ρ_t describes their positions at time t.

Example: $v(x,t) = -x$ and $Z_0 \sim N(0,1)$.

$$\frac{dZ_t}{dt} = -Z_t \quad \Longrightarrow \quad Z_t = e^{-t} Z_0.$$

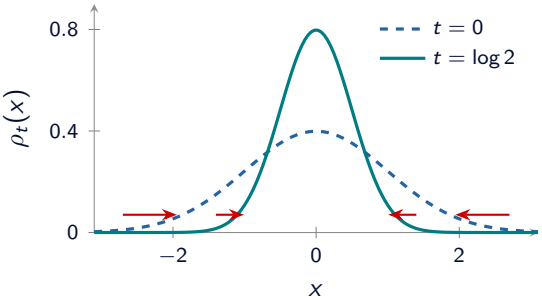
Thus

$$\boxed{Z_t \sim N(0, e^{-2t}).}$$

At $t = \log 2$, every position is halved:

$$Z_{\log 2} = \tfrac{1}{2} Z_0.$$

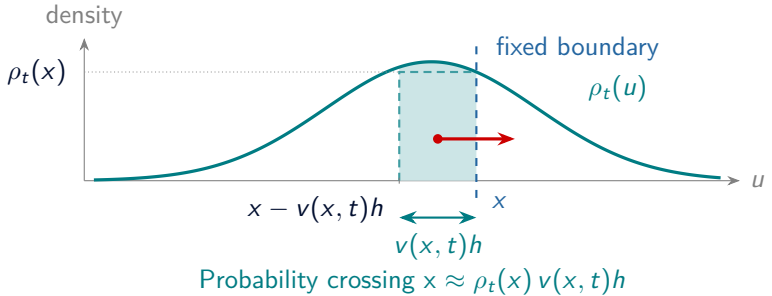
The density becomes **narrower and taller**.



- Each path is **deterministic** once Z_0 is given. The randomness comes from Z_0.

Probability flux equals velocity times density

Fix a location x and suppose $v(x,t) > 0$. Over a short time h, particles near x move about $v(x,t)h$ to the right.



Recall that **probability flux** at x is defined as the **net rate of probability flow across** x:

$$\boxed{J_t^{\mathrm{ODE}}(x) = v(x,t)\rho_t(x).}$$

If $v(x,t) < 0$, the flux is negative: probability flows to the left.

Matching the SDE probability flux

Recall the forward SDE and its marginal density:

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t, \qquad X_t \sim q_t.$$

Its Fokker–Planck equation has flux

$$\partial_t q_t = -\partial_x J_t, \qquad J_t = f(x,t)q_t(x) - \frac{1}{2}g^2(t)\partial_x q_t(x).$$

Match the fluxes: $J_t^{\mathrm{ODE}}(x) = v(x,t)q_t(x) = J_t(x)$. Dividing by $q_t(x) > 0$ gives

$$\boxed{v(x,t) = \frac{J_t(x)}{q_t(x)} = f(x,t) - \frac{1}{2}g^2(t)\underbrace{\partial_x \log q_t(x)}_{s_t(x)}.}$$

The resulting **probability flow ODE** is

$$\boxed{dZ_t = \left[f(Z_t,t) - \frac{1}{2}g^2(t)s_t(Z_t)\right]dt, \qquad Z_0 \sim q_0.}$$

The score correction turns the diffusion flux into deterministic transport.

The probability flow ODE

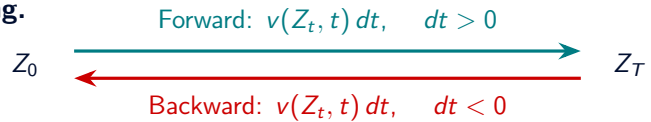
Matching the marginals. For the SDE solution $X_t \sim q_t$, the **probability flow ODE** is

$$dZ_t = \left[f(Z_t, t) - \frac{1}{2} g^2(t) s_t(Z_t) \right] dt, \qquad Z_0 \sim q_0.$$

Same probability flux and **same initial density** give, under uniqueness,

$$\boxed{Z_t \sim q_t, \qquad 0 \le t \le T.}$$

Backward sampling.



Same trajectory, opposite time directions.

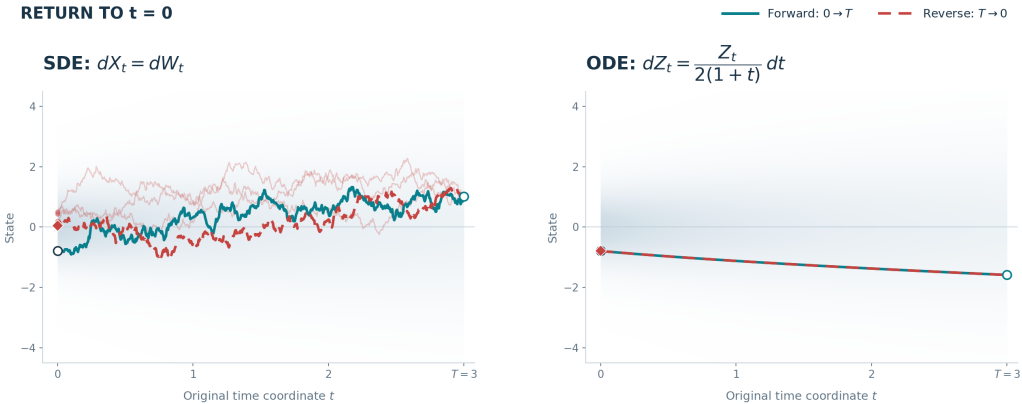
Draw $Z_T \sim q_T$ and integrate the **same ODE** from T to 0:

$$dZ_t = \left[f(Z_t, t) - \frac{1}{2} g^2(t) s_t(Z_t) \right] dt, \qquad dt < 0.$$

Result: $Z_0 \sim q_0$.

SDE and ODE roundtrip

$X_0 = Z_0 \sim N(0, 1)$ $q_t = N(0, 1 + t)$ $T = 3$



Open animation in browser

The **ODE retraces its path**, while the **reverse SDE uses fresh noise** to generate a new path with the same marginals.

Training the score network

Fix the **forward SDE** and learn its **time-dependent score**:

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t, \qquad s_\theta(x, t) \approx \nabla_x \log q_t(x).$$

Denoising score matching. Sample

$$t \sim U(0, T), \qquad X_0 \sim q_0, \qquad X_t \sim q_{0t}(\cdot \mid X_0).$$

Then minimize Song et al.'s objective:

$$\boxed{\theta^* = \arg\min_\theta \mathbb{E}_t \Big\{ \lambda(t)\, \mathbb{E}_{X_0} \mathbb{E}_{X_t \mid X_0} \Big[\| s_\theta(X_t, t) - \nabla_{X_t} \log q_{0t}(X_t \mid X_0) \|_2^2 \Big] \Big\}.}$$

Conditional-score targets teach the marginal score.

Gaussian case (VP/VE): the training target is explicit:

$$X_t = \alpha_t X_0 + \sigma_t \epsilon, \quad \epsilon \sim N(0, I_D) \quad \Longrightarrow \quad \nabla_{X_t} \log q_{0t}(X_t \mid X_0) = -\frac{\epsilon}{\sigma_t}.$$

With $\lambda(t) = \sigma_t^2$, this is equivalent to the objective for **DDPM/SMLD**.

SDE and ODE sampling

Initialize X_T, Z_T from noise. Step backward from t to $t - h$ using the **same trained score**.

Reverse SDE: Euler–Maruyama

$$X_{t-h} \approx X_t + \left[-f(X_t, t) + g^2(t) s_\theta(X_t, t)\right] h + g(t)\sqrt{h}\,\xi, \qquad \xi \sim N(0, 1).$$

Probability flow ODE: Euler

$$Z_{t-h} \approx Z_t + \left[-f(Z_t, t) + \frac{1}{2} g^2(t) s_\theta(Z_t, t)\right] h.$$

Reevaluate the score at each new state and time; repeat until $t = 0$.

- **SDE:** Sample fresh, independent noise ξ at each step.
- **ODE:** With Z_T and deterministic solver settings fixed, the output is reproducible.
- Higher-order or adaptive ODE solvers can improve the trade-off between accuracy and computational cost.

Probability-flow ODEs to flow matching

A velocity field determines how samples and their density evolve:

$$\frac{\mathrm{d}Z_t}{\mathrm{d}t} = v(Z_t, t), \qquad \frac{\partial q_t(x)}{\partial t} = -\nabla_x \cdot \big[q_t(x) v(x,t)\big].$$

Probability-flow ODE

Learn the **score**: $s_\theta(x,t) \approx \nabla_x \log q_t(x)$.

Then construct the velocity:

$$v_\theta(x,t) = f(x,t) - \tfrac{1}{2} g(t)^2 s_\theta(x,t).$$

Flow matching

Choose a probability path q_t between noise and data.

Learn a **velocity field directly**:

$$v_\theta(x,t) \approx v(x,t).$$

Next lecture: how do we learn the velocity?

How can noise and data samples provide training targets for $v_\theta(x,t)$?

References

- **Särkkä, S. and Solin, A. (2019).**
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  Reverse-time diffusion equation models.
  Stochastic Processes and their Applications, 12(3), 313–326.

- **Song, Y., et al. (2021).**
  Score-Based Generative Modeling through Stochastic Differential Equations.
  International Conference on Learning Representations (ICLR).